

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

**JANSETU: An Intelligent Complaint Prioritization and
Analytics-Based Routing System for Government Grievance
Portals**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Title of the Project**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gifty Gupta**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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1. Abstract

The Jansetu system is a web-based complaint management platform designed to digitize and streamline the process of grievance registration, tracking, and resolution. Traditional complaint systems often suffer from delays, lack of transparency, and inefficient tracking mechanisms. Jansetu addresses these issues by providing a centralized and structured digital platform.

The system allows users to easily register complaints along with relevant details, while administrators can efficiently manage, categorize, and update the status of each complaint. A role-based access system ensures secure separation between users and administrators. Additionally, automated workflows improve processing speed and reduce manual effort.

Overall, Jansetu enhances transparency, accountability, and efficiency in complaint handling, making grievance redressal faster and more organized.

2. Introduction

In traditional complaint handling systems, grievances are often recorded manually or through unstructured processes, leading to delays, lack of tracking, and poor transparency. Users face difficulties in knowing the status of their complaints, while administrators struggle with organizing and prioritizing large volumes of requests.

To overcome these challenges, Jansetu is introduced as a web-based complaint management system. It provides a centralized platform where complaints can be registered, tracked, and resolved efficiently.

The system is designed with separate user and admin roles. Users can submit complaints with relevant details, while administrators can manage, categorize, and update their status. This improves accountability, reduces processing time, and ensures a more structured grievance redressal process.

3. Methodology

3.1 System Architecture

- Jansetu is built using a client-server architecture
- The system is divided into three main layers: frontend, backend, and database
- Ensures modularity, scalability, and easy maintenance

3.2 Frontend Design

- Developed using HTML, CSS, and React.js
- Provides user-friendly interface for users and admin
- Handles user interactions like login, registration, and complaint submission
- Responsive design ensures accessibility across devices

3.3. Backend Processing

- Built using Node.js / Java (based on implementation)
- Handles authentication, API requests, and complaint processing
- Implements business logic such as status updates and priority assignment
- Ensures secure and efficient data handling

3.4 Database Management

- Uses MySQL database for structured data storage
- Stores user details, complaint records, and status history
- Ensures data consistency and integrity
- Supports fast retrieval and updates

3.5 Workflow of the System

- User registers and logs into the system
- Complaint is submitted through a form with details
- Backend validates and stores complaint in database
- Admin reviews complaint from dashboard
- Priority is assigned based on severity
- Status is updated until final resolution

4. Tools and Technologies

4.1 Frontend Development

- React.js used for building dynamic user interface
- HTML5 used for structuring web pages
- CSS3 used for styling and layout design
- JavaScript used for client-side interactions

4.2 Backend Development

- Node.js / Java used for server-side development
- Express.js used for API creation and routing (if Node.js)
- REST APIs used for communication between frontend and backend

4.3 Database

- MySQL used as relational database management system
- Stores user details, complaints, and status records
- Ensures structured and secure data storage

4.4. Security & Authentication

- JWT (JSON Web Token) used for authentication
- Password encryption for secure login system
- Role-based access control for user and admin separation

4.5 Development & Testing Tools

- Visual Studio Code used as code editor
- Postman used for API testing
- Git & GitHub used for version control

5. Implementation

5.1 User Module Implementation

- Users can register and create personal accounts
- Secure login system implemented for authentication
- Complaint submission form allows users to enter issue details
- Users can view and track complaint status in real time

5.2 Admin Module Implementation

- Admin dashboard implemented for managing all complaints
- Admin can view, filter, and categorize complaints
- Status updates (Pending / In Progress / Resolved) are managed by admin
- Priority assignment is handled based on complaint type and urgency

5.3 Complaint Handling System

- Each complaint is assigned a unique ID automatically
- Complaints are stored in database after submission
- Admin reviews and updates status through dashboard
- Changes are reflected instantly for users

5.4 Data Flow Implementation

- Frontend sends complaint data to backend via API
- Backend validates and processes incoming requests
- Data is stored in MySQL database
- Updated status is fetched and displayed to user interface

5.5 Security Implementation

- JWT-based authentication used for secure sessions
- Role-based access ensures separation of user and admin data
- Passwords are stored in encrypted format

6. Outcomes

6.1 System Performance Analysis

- The system demonstrates fast response time for complaint submission and retrieval
- Backend processing ensures smooth handling of multiple user requests simultaneously
- Database operations (insert, update, fetch) are executed efficiently
- System remains stable under normal usage conditions

6.2 User-Side Outcomes

- Users can successfully register and authenticate without errors
- Complaint submission process is simple and structured
- Users are able to track complaint status at any time
- Clear visibility of complaint progress improves user satisfaction

6.3 Admin-Side Outcomes

- Admin dashboard provides centralized control over all complaints
- Complaints can be filtered based on category, status, and priority
- Admin can update complaint status in real time
- Priority-based sorting helps in handling urgent issues first

6.4 Functional Effectiveness

- Each complaint is assigned a unique identifier for tracking
- Automated workflow reduces manual intervention in processing
- Role-based access ensures proper separation of responsibilities
- System prevents unauthorized access to sensitive operations

6.5 Efficiency and Improvements Achieved

- Significant reduction in manual paperwork and record handling
- Faster grievance resolution compared to traditional methods
- Improved communication flow between user and administration
- Structured categorization reduces confusion in complaint handling

6.6 Overall System Impact

- Jansetu provides a centralized and organized complaint management solution
- Enhances transparency by allowing users to track complaint status
- Improves accountability of administrative processes
- Ensures a more reliable and scalable grievance redressal system

7. Conclusion and Scope

7.1 Conclusion

- Jansetu provides a centralized and efficient complaint management system
- It replaces traditional manual grievance handling with a structured digital approach
- The system ensures transparency in complaint tracking and resolution process
- Role-based access improves security and proper distribution of responsibilities
- Overall, the system enhances speed, accuracy, and reliability in grievance redressal

7.2 Future Scope

- Development of a mobile application for improved accessibility
- Integration of AI-based complaint categorization for automated sorting
- Implementation of WhatsApp/SMS notifications for real-time updates
- Addition of multilingual support for wider user accessibility
- AI chatbot integration to assist users in complaint registration and tracking
- Advanced analytics dashboard for monitoring complaint trends and performance
- Integration with government portals for broader implementation and impact

8. References

8.1 Documentation Sources

- React.js Official Documentation – <https://react.dev/>
- Node.js Official Documentation – <https://nodejs.org/>
- Express.js Framework Guide – <https://expressjs.com/>
- MySQL Official Documentation – <https://dev.mysql.com/doc/>

8.2 Development References

- JWT Authentication Guide – <https://jwt.io/introduction>
- MDN Web Docs (HTML, CSS, JavaScript) – <https://developer.mozilla.org/>

8.3 Project Support Materials

- Stack Overflow – <https://stackoverflow.com/> (for technical problem solving)
- GitHub Repositories – <https://github.com/> (for reference implementations)

9. Appendices

9.1 Database Structure

- The system uses a relational database (MySQL) for data storage
- User Table includes fields such as user_id, name, email, password, and role
- Complaint Table includes complaint_id, user_id, title, description, category, status, and timestamp
- Each complaint is linked to a user using a foreign key relationship (user_id)
- Status field tracks complaint progress such as Pending, In Progress, and Resolved

9.2 API Endpoints

- POST /register – Used for user registration
- POST /login – Used for user authentication
- POST /complaint – Used to submit a new complaint
- GET /complaint – Used to retrieve complaint details
- PUT /complaint/:id – Used to update complaint status
- GET /user/complaints – Used to fetch complaints specific to a user